

AirOne — Documentation

This directory is the complete operator- and engineer-facing documentation set for the AirOne CanSat Ground Station & Scientific Analysis Platform.

Everything here reflects the **actual behaviour of the code in this repository** — no aspirational features are documented. Where a capability is optional or can degrade, the documentation states the exact condition and the observable result.

Contents

Document	Audience	Purpose
operator_manual.md	Mission operators	Day-of-mission operation, GUI, judge/presentation mode
engineering_manual.md	Developers / integrators	Architecture, workers, capability tiers, extending the system
api_reference.md	API consumers	Every REST endpoint, roles, request/response envelopes
telemetry_protocol.md	Firmware / RF engineers	Wire format, CRC, FEC, sequence handling
scientific_methods.md	Scientists / judges	The 37 analyses, their equations, assumptions, and uncertainty
ml_tier.md	Data scientists	Anomaly detectors, drift, model provenance, advisory-only guarantee
failure_modes.md	Everyone	How every subsystem degrades honestly (the quality/state vocabulary)
security_model.md	Security reviewers	Accounts & logout, JWT, RBAC, telemetry HMAC link

Document	Audience	Purpose
configuration.md	Operators / DevOps	auth, audit chain, secret handling Every config key and its effect
troubleshooting.md	Everyone	Symptoms → diagnosis → resolution
test_plan.md	QA / judges	What the 154-test suite proves and how to run it

The one rule that governs everything

Never present fabricated, stale, or fake-complete data as real.

Every subsystem either (a) works completely, (b) is explicitly disabled with a visible reason, or (c) is unavailable because of a genuinely missing dependency that is reported with a clear diagnostic. When data quality is uncertain, the system says so using an explicit vocabulary (VALID, SUSPECT, INVALID, MISSING, STALE, CORRUPTED, SIMULATED, ESTIMATED, DEGRADED, NOT_CONFIGURED, UNAVAILABLE). See [failure_modes.md](#).